

MEASURING AI READINESS OF SEMANTIC MODELS

Is your legacy of 10+ years of semantic models **ready for AI?**

AI-READY SEMANTIC MODELS

- ✓ Clear Names
- ✓ Rich Descriptions
- ✓ Well-Defined Measures
- ✓ Strong Relationships
- ✓ Business Friendly

I know what this means...
but will **AI?**

AI READINESS SCORE



- ✓ Metadata
- ✓ Model Structure
- ✓ Measures
- ✓ Relationships
- ✓ Discoverability

Notebook

```
1 # Assess AI readiness
2 import sempy_labs as labs
3 from semantic_link import ....
4
5 # Scan semantic models
6 models = labs.list_semantic_models()
7
8 # Evaluate readiness
9 score = labs.evaluate_ai_readiness(models)
10
11 # Show results
12 display(score)
```



Semantic Link (Labs)

- ✓ Inspect
- ✓ Analyze
- ✓ Assess
- ✓ At Scale



Microsoft
Fabric



Notebooks



Semantic Link



Semantic Link
(Labs)



Insights at
Scale

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Learning objectives

Readiness

Identify the key characteristics of an AI-ready Power BI semantic model

Assess

Assess the AI readiness of Power BI semantic models programmatically using Semantic Link and Semantic Link Labs

Monitor

Monitor and compare AI readiness across a Power BI landscape, enabling organizations to prioritize improvements

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 Data-Marc.com

 DutchFabricUsergroup.com

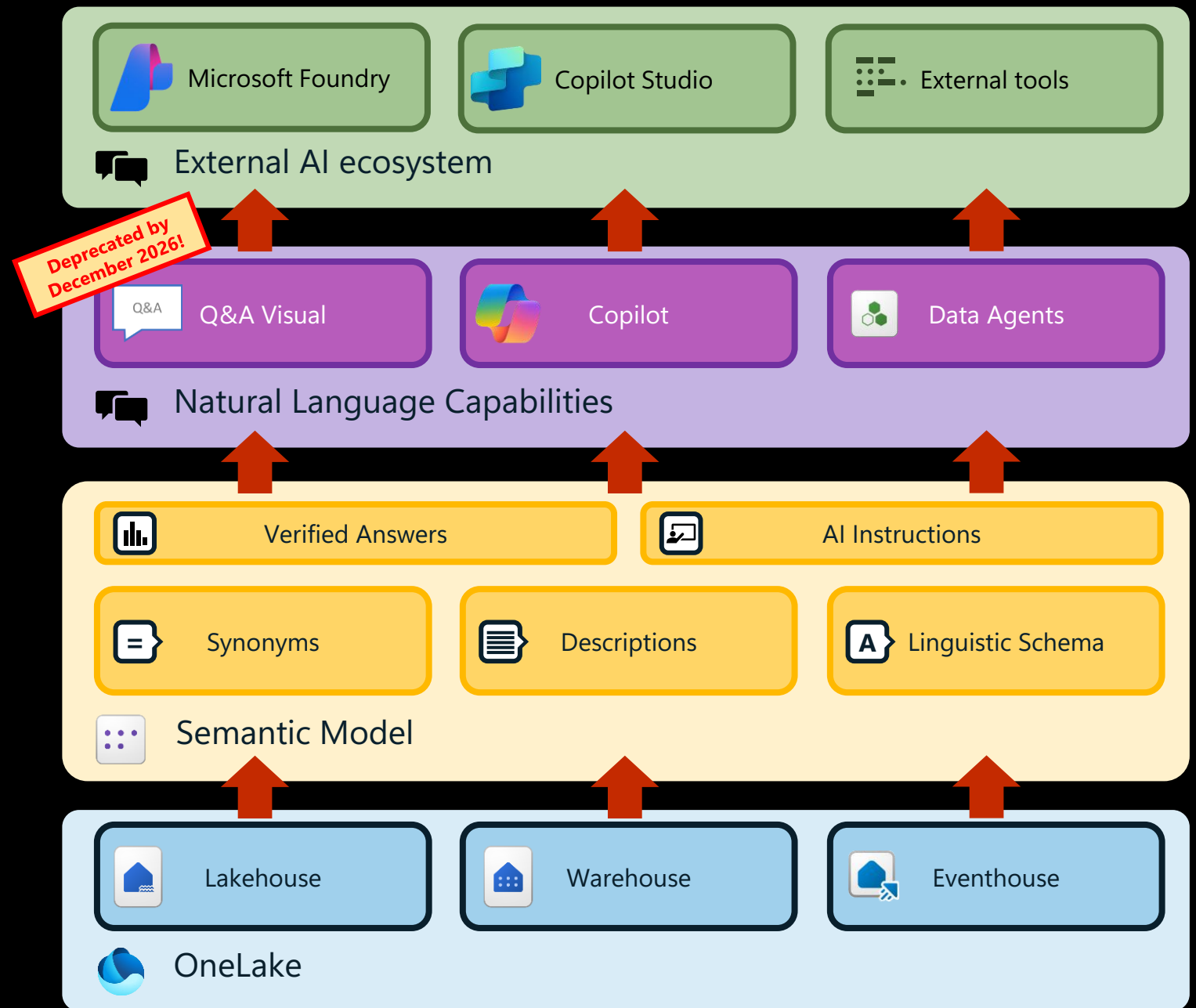
FAVORITE STUFF:





The AI ready Semantic Model

Why Semantic Models matter for AI experiences

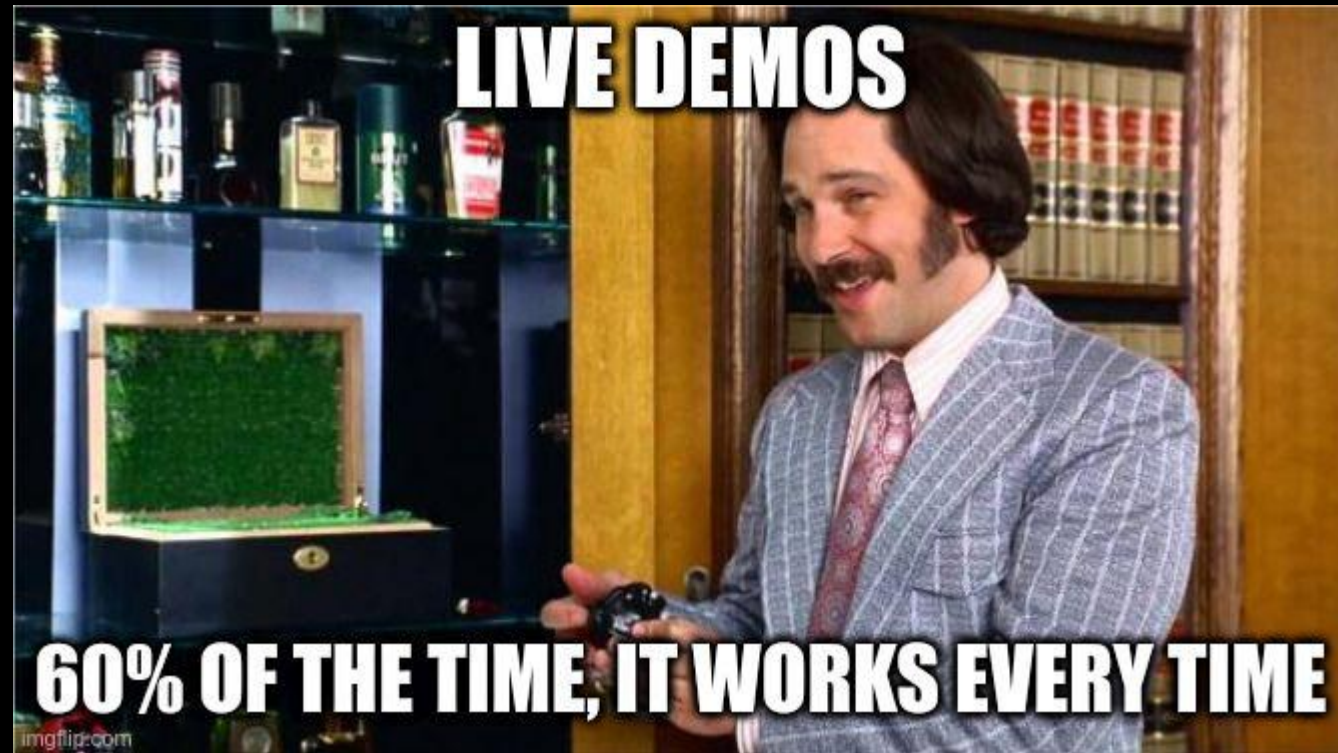


Facing the problem

We have 10+ years of legacy Power BI Semantic Models, which were build in times before AI.

Things that matter today for Large Language Models, were not prioritized in the past!

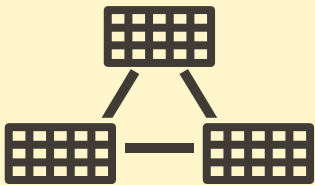
Demo



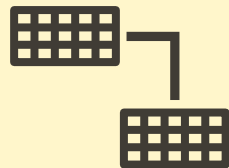
Modeling fundamentals

A **good** semantic model, covers at least the following items:

Star schema



Relationships



Object metadata



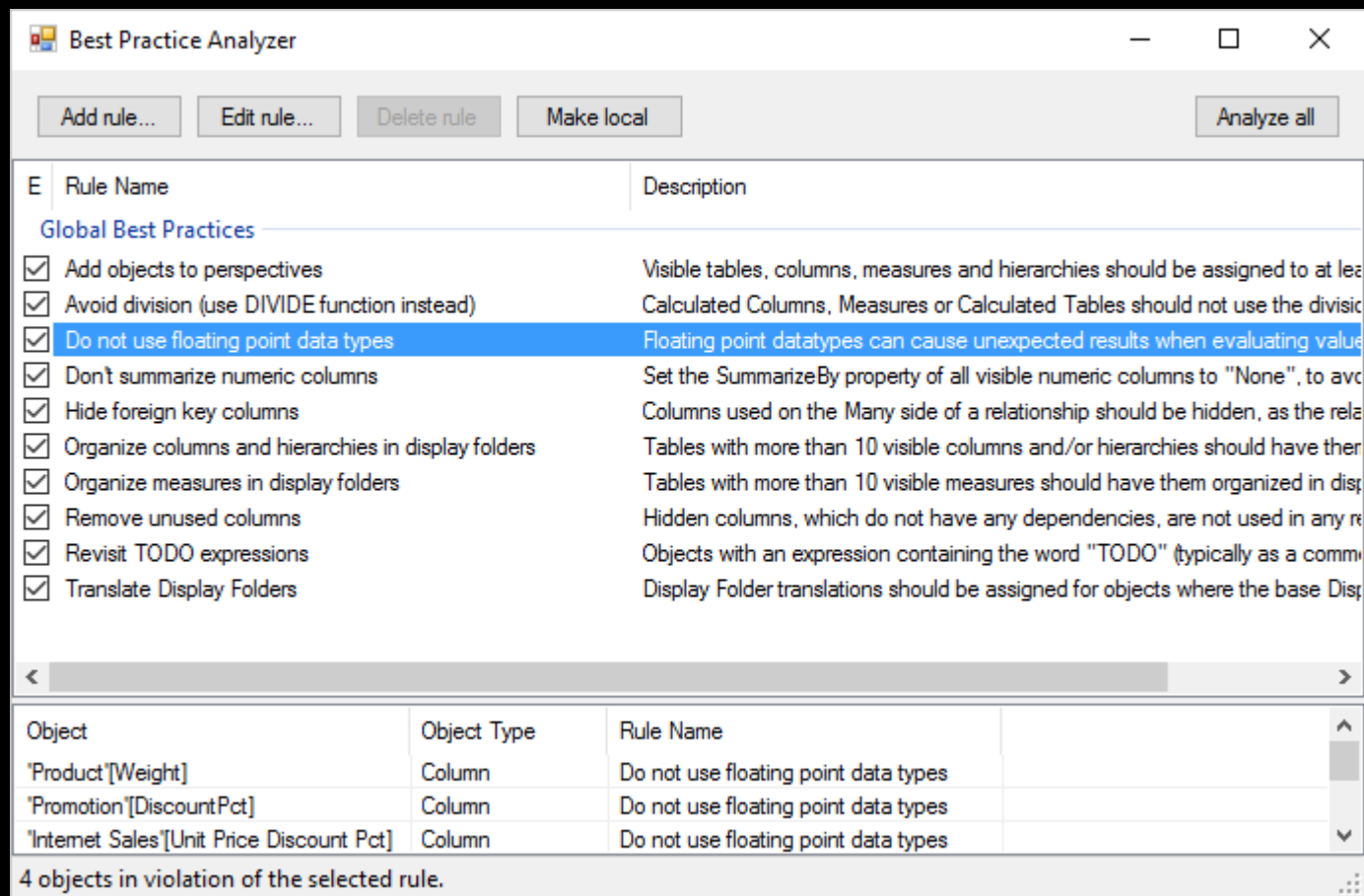
Measures



Hide & see



Tabular Editor Best Practices Analyzer



Best Practice Analyzer

Buttons: Add rule..., Edit rule..., Delete rule, Make local, Analyze all

E	Rule Name	Description
Global Best Practices		
☒	Add objects to perspectives	Visible tables, columns, measures and hierarchies should be assigned to at least one perspective.
☒	Avoid division (use DIVIDE function instead)	Calculated Columns, Measures or Calculated Tables should not use the division operator (/).
☒	Do not use floating point data types	Floating point datatypes can cause unexpected results when evaluating values.
☒	Don't summarize numeric columns	Set the SummarizeBy property of all visible numeric columns to "None", to avoid summarizing them.
☒	Hide foreign key columns	Columns used on the Many side of a relationship should be hidden, as the relationship is not visible.
☒	Organize columns and hierarchies in display folders	Tables with more than 10 visible columns and/or hierarchies should have them organized in display folders.
☒	Organize measures in display folders	Tables with more than 10 visible measures should have them organized in display folders.
☒	Remove unused columns	Hidden columns, which do not have any dependencies, are not used in any report.
☒	Revisit TODO expressions	Objects with an expression containing the word "TODO" (typically as a comment) should be revisited.
☒	Translate Display Folders	Display Folder translations should be assigned for objects where the base Display Folder is not English.

4 objects in violation of the selected rule.

Object	Object Type	Rule Name
'Product'[Weight]	Column	Do not use floating point data types
'Promotion'[DiscountPct]	Column	Do not use floating point data types
'Internet Sales'[Unit Price Discount Pct]	Column	Do not use floating point data types

Humans vs AI

The gap between a model that works for humans and one that works well for AI

Semantic Model design:

- **Human interpretation:** Cust, Prod, FactSales, DimDate
- **Context around measures:** Sales, is it with or without VAT? And which measure represents Sales? Revenue, Gross sales, Net Sales, ...
- **Ambiguity:** “Show our best customers” what is best?
- **Relationship implicitly:** [Customer] 1:N [Sales] Sales belongs to a customer but might be modeled differently. Inactive relationships, many-to-many, multiple date relationships (orderdate, shipdate, ...)

AI challenge:

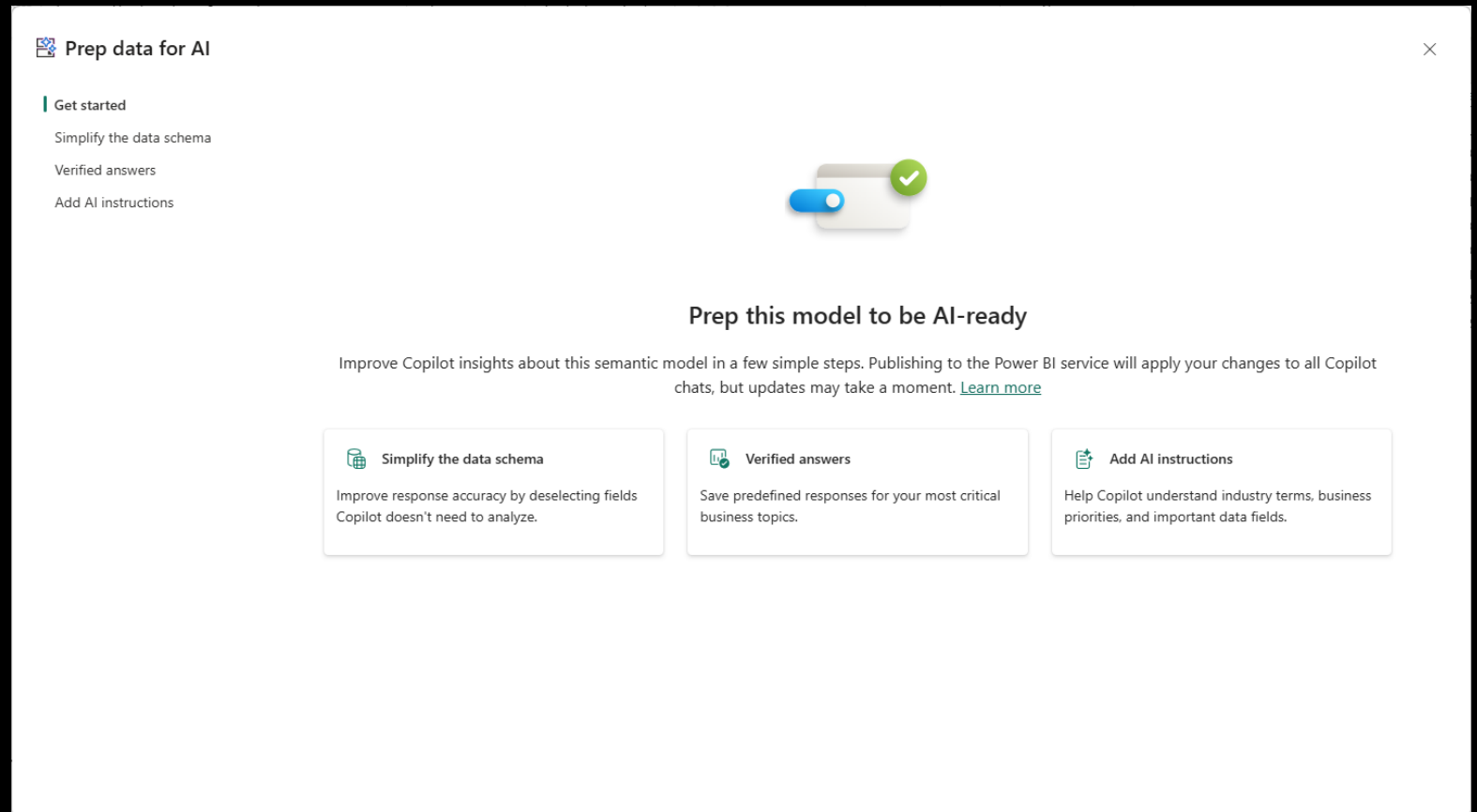
- Business meaning is not always explicitly represented in the model.
- AI must infer meaning from names, descriptions, metadata, relationships and calculations.
- The less explicit the semantics, the more room there is for incorrect interpretation.
- Unable to use features like Visual Calculations, Field parameters and more... (Report design related features)

What makes a Semantic Model **AI ready**?

Prep data for AI!

Three main pillars:

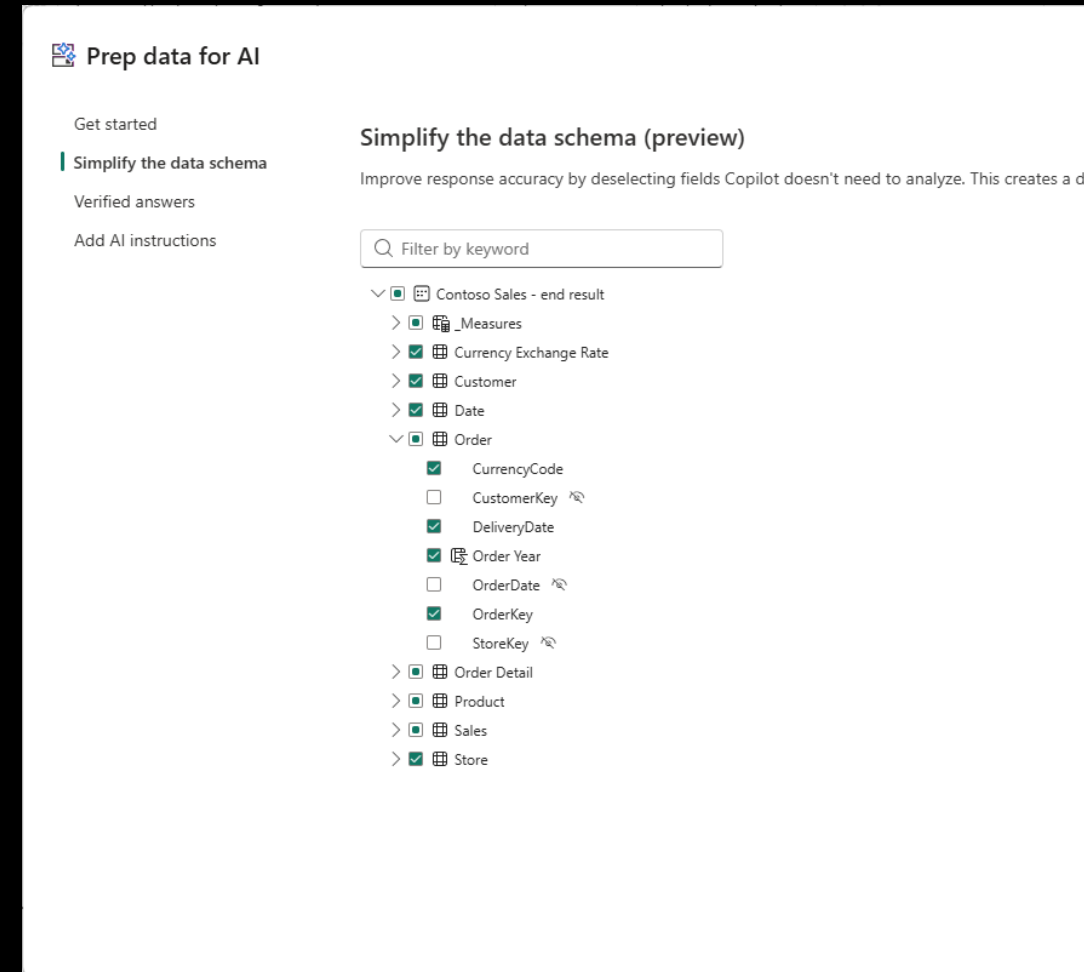
- Simplify schema
- Verified answers
(links to visuals)
- AI Instructions



Prep data for AI: **Simplify schema**

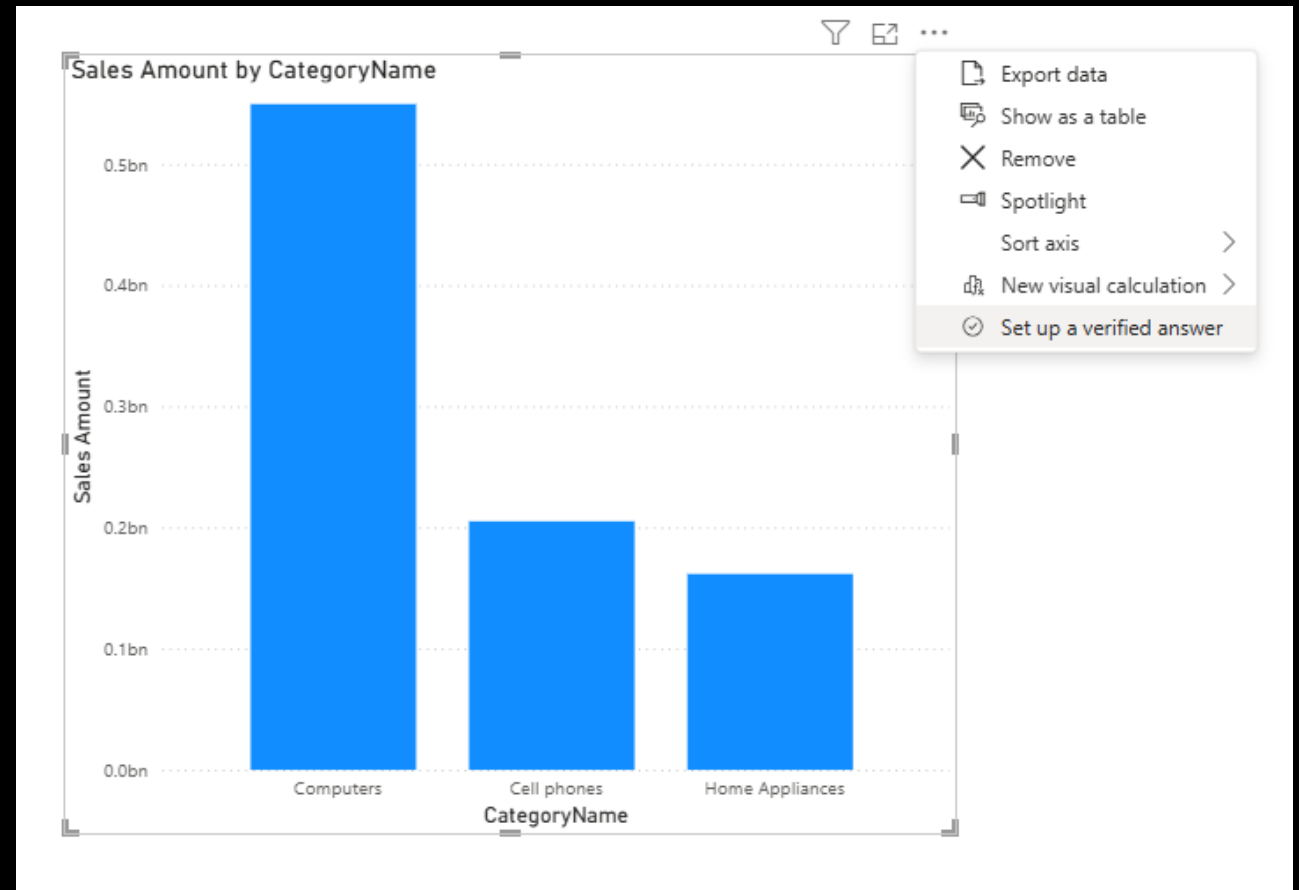
- Only keep what is relevant!
- Hide report-specific measures
 - Conditional formatting
 - Measures outputting texts like visual titles
- Disconnected tables likely cause issues
→ build relationships!

Keep in mind that Visual Calculations are not part of the semantic model!



Prep data for AI: **Verified answers**

- Verified answers will always be referenced first
- Selected on visual level
- Linked to specific prompts
 - Copilot suggestions for relevant prompts
 - Ability to add your own prompts



Prep data for AI: Verified answers

Prep data for AI

Get started

Simplify the data schema

Verified answers

Add AI instructions

Verified answers (preview)

← All verified answers

Phrases connected to verified answers

Add phrases people might use when asking about the model data, and Copilot will respond with this visual. Test your phrases out in the chat before publishing.

What are my top 3 performing product categories

Add

☑ Show the top 3 categories by sales.

Copilot suggestions

Which product categories have the highest sales?

What are sales by category?

Refresh

Copilot uses AI. Always review content for mistakes. [Read terms](#)

Visual

The visual below will appear in Copilot's responses to phrases you add. Any changes to the report won't impact this visual.

CategoryName	Sales Amount
Computers	0.55bn
Cell phones	0.2bn
Home Appliances	0.15bn

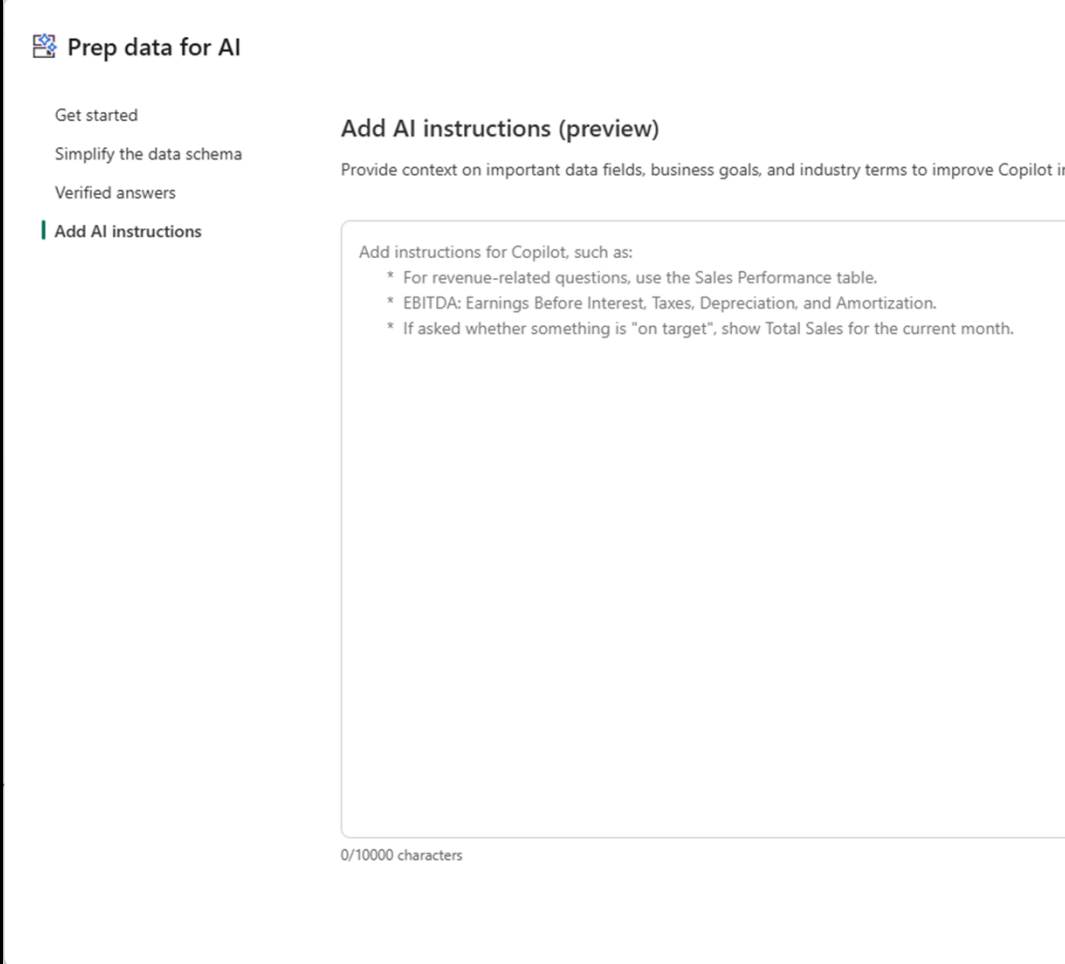
Apply

Close

 Data-Marc.com

Prep data for AI: **AI Instructions**

- Unstructured guidance (textual)
- No guarantee the guidance is followed exactly – see it as a hint!
- Response formatting
 - Style
 - Tone
- Business specific logic – for example:
 - Define what is a top performer – high or low?
 - Peak / off-season: which months are these?



The screenshot shows a web interface titled "Prep data for AI". On the left is a sidebar with four items: "Get started", "Simplify the data schema", "Verified answers", and "Add AI instructions" (which is highlighted with a green bar). The main content area is titled "Add AI instructions (preview)" and includes a sub-header "Provide context on important data fields, business goals, and industry terms to improve Copilot in". Below this is a large text input area with the prompt "Add instructions for Copilot, such as:" followed by three bullet points: "For revenue-related questions, use the Sales Performance table.", "EBITDA: Earnings Before Interest, Taxes, Depreciation, and Amortization.", and "If asked whether something is 'on target', show Total Sales for the current month." At the bottom right of the input area is a character count "0/10000 characters".

Prep data for AI

Get started
Simplify the data schema
Verified answers
Add AI instructions

Add AI instructions (preview)

Provide context on important data fields, business goals, and industry terms to improve Copilot in

Add instructions for Copilot, such as:

- * For revenue-related questions, use the Sales Performance table.
- * EBITDA: Earnings Before Interest, Taxes, Depreciation, and Amortization.
- * If asked whether something is "on target", show Total Sales for the current month.

0/10000 characters



Measuring AI readiness

Turning it into a measurable criteria

Introducing the **AI readiness score** for Semantic Models!

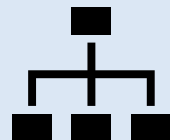
Discoverability & Metadata



Can AI discover and understand what the model contains?

Max score: 20

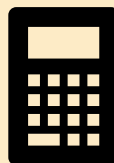
Model Structure & Organization



Is the model organized in a predictable way?

Max score: 20

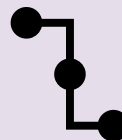
Measures & Calculations



Can AI reliably understand and use the calculations?

Max score: 20

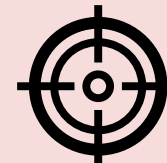
Relationships & Model Logic



Can AI determine how entities relate and how filters flow?

Max score: 20

Business Semantics & Context



Does the model express business meaning not just technical structure?

Max score: 10

Quality & Trust



Are there structural (data-quality) issues that could influence AI answer quality?

Max score: 10

Discoverability & Metadata



Can AI discover and understand what the model contains?

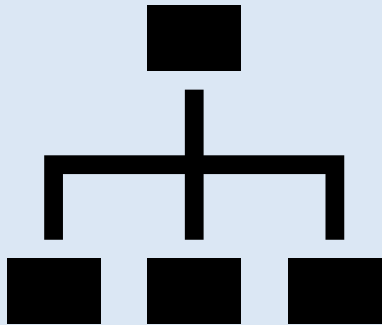
Max score: 20

What	Points	How to measure
Table descriptions	3	% of visible tables with a non-empty description
Column descriptions	4	% of relevant columns with a non-empty description
Measure descriptions	5	% of measures with a non-empty description
Business-friendly names	4	% of objects following naming rules / avoiding technical names and abbreviations
Synonyms defined	4	% of business facing (non-hidden) objects having synonyms defined

Metadata score =

*Table coverage
+ column coverage
+ Measure coverage
+ naming quality
+ synonym coverage*

Model Structure & Organization



Is the model organized in a predictable way?

Max score: 20

What	Points	How to measure
Star schema characteristics	5	Dimensions & facts follow defined
Date Table is flagged as such	4	Is there a Date table present and flagged as such?
Facts & dimensions can be identified	3	Tables can be unambiguously identified, and other irrelevant tables are hidden
Technical tables are hidden (for AI)	4	Technical / helper tables are hidden for AI.
Auto summarization for numeric columns is set	4	% of numeric columns used in relationships or measures having auto summarization configured

Model structure score=

Star schema

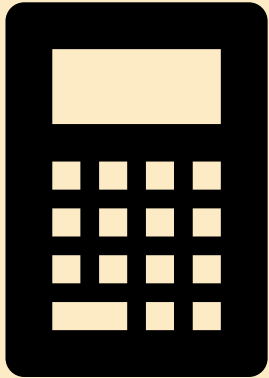
+ Date table availability

+ Identifiable facts & dimension

+ Hidden technical items

+ Auto summarization config

Measures & Calculations



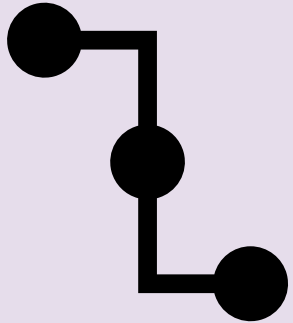
Can AI reliably understand
and use the calculations?

Max score: 20

What	Points	How to measure
Measures clearly named	5	Naming follows semantic/business naming conventions
Measures have descriptions	5	% description coverage
Format strings are applied	4	% measure having format strings applied
Time intelligence available	4	Typical patterns like YTD / LY / SPLY are available in the model
Measures are organized	2	Grouped together logically (with display folders)

Measure & calculations score=
Measure naming
+ Measure Descriptions
+ Measure with Format Strings
+ Time intelligence
+ Measure organization

Relationships & Model Logic



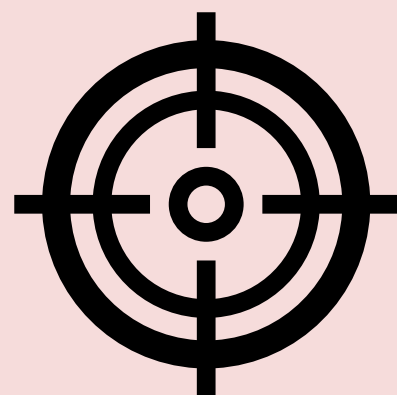
Can AI determine how entities relate and how filters flow?

Max score: 20

What	Points	How to measure
Appropriate active relationships	4	% of expected relationships are set to active
Unambiguous filter paths	3	Detect ambiguous paths
Correct cardinality	6	1:N / 1:1 relationships only, bi-directionals should be avoided
Avoid unnecessary bi-directional filter paths	4	Avoid bi-directional filter paths
Relationships are documented (have a description)	3	% Relationship metadata availability

Relationships & Model Logic score=
Active relationships
+ Unambiguous paths
+ Cardinality
+ Bi-directional filter paths
+ Relationship documentation

Business Semantics & Context



Does the model express business meaning not just technical structure?

Max score: 10

What	Points	How to measure
AI Instructions / Notes for AI	5	Context and business specifics are described in the AI Instructions
Calculation groups used	2	Repeated (time intelligence / currency / ...) calculation patterns covered in Calculation groups
Business context modelled in hierarchies	1	Logical drill-down structures
Units, currency & formatting defined	2	% / currency / quantity / decimal places / date formatting set for measures and numeric columns

Business Semantics & Context score=
AI Instructions
+ Calculation groups
+ Hieraries
+ Formatting

Quality & Trust



Are there structural (data-quality) issues that could influence AI answer quality?

Max score: 10

What	Points	How to measure
No columns with solely the same value or empty	3	Validate availability of data and cardinality of columns
Data types are set consistency on both ends of the relationship	2	Inconsistent datatypes will make relationships unusable
No duplicate measures	2	Different measures with same definition will result in score 0
Security roles configured	2	Security roles created and have expressions added
Security roles documented	1	Descriptions on roles added

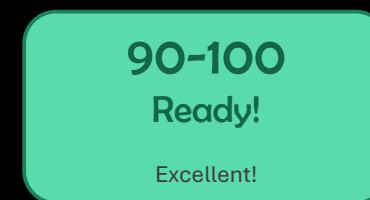
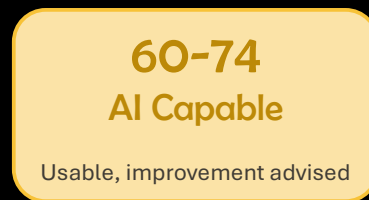
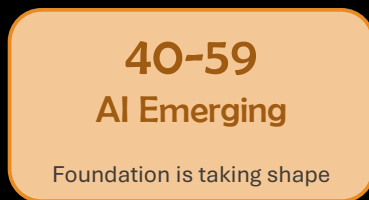
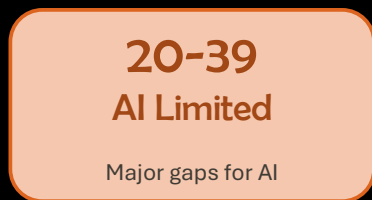
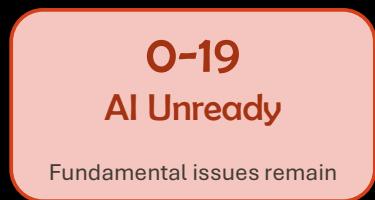
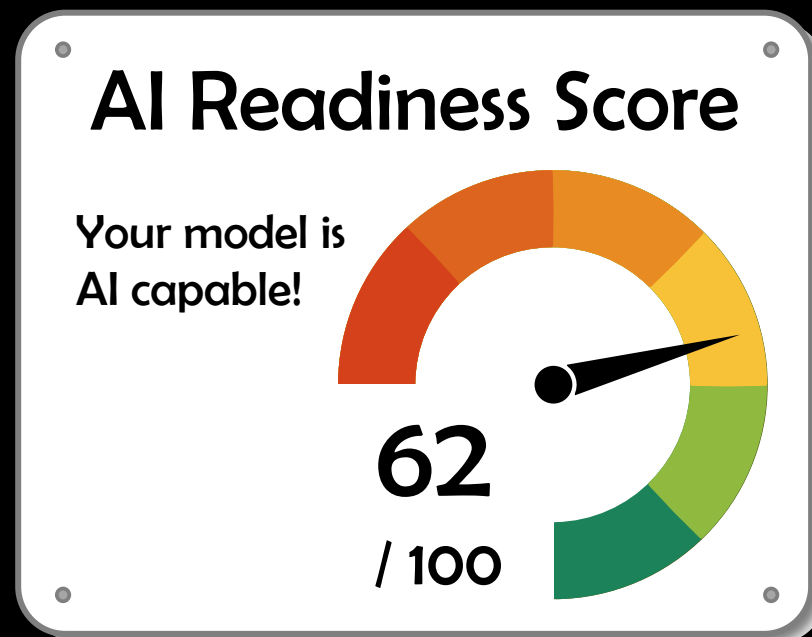
Quality & Trust score=

*Columns with one value
+ Data types across relationships
+ Duplicate measures
+ Security roles configured
+ Security roles documented*

The AI Readiness Assessment

A semantic model can be perfectly usable for humans while still leaving important context implicit for AI.

The AI Readiness Score makes these gaps **visible, measurable, and actionable.**



Common issues

- Conflicting or overly complex instructions
- Skipping schema refinement – especially with large semantic models containing many tables
- Not using a star schema (resulting in inefficient DAX)
- Relying on hidden fields – copilot cannot access them
- Including unnecessary measures: Simplify the schema,
 - Remove measures that are not relevant or broken,
 - Hide report measure (for conditional formatting for example),
 - Avoid duplicating logic in different measures
- Non-descriptive naming (technical names for objects)
- Relying on implicit measures: high chance of inconsistency!
- Ambiguous date fields (e.g. Order Date, Ship Date, ...): use prep data for AI to specify when to use which



Programmatically inspect Semantic Models

Why manual assessments don't scale?!

Manual assessment	The problem
Open the semantic model manually	Repeat for every semantic model in scope
Check tables & columns	Lots of meta data to process
Review measures	Business logic is harder to assess though, technical assessment only cannot do the job either
Check names & synonyms	Subjective and inconsistent
Record findings	Documentation quickly becomes outdated
Repeatability after every change	Continuous maintenance becomes impossible
Self-service users should be able to run the assessment themselves	Requires understanding across all users

The result?

Slow, subjective, inconsistent and not easily repeatable

Bringing in Semantic Link (Labs)

Instead of asking “is this model AI-ready?”
we can start measuring the characteristics that make it IT-ready!

Semantic Link

- Python library for interacting with semantic models
- Used inside of a Fabric notebook
- Pre-installed in the Fabric runtime
- Supported by Microsoft

Semantic Link Labs

- Open-source extension of Semantic Link
- Higher level inspection functions
- Model management capabilities

What is Semantic Link?

- Python library for interacting with semantic models
- Used inside of a Fabric notebook
- Pre-installed in the Fabric runtime
- Supported by Microsoft
- List_* functions to view semantic model metadata
- Functions to execute DAX, TMSL, XMLA, DMVs
- Basic API wrapper functions (list_capacities, list_workspaces, list_reports, list_items)
- And more...

Semantic Link – Use cases

Power BI general

- Documenting Power BI items
- Move Power BI items across workspaces
- Detect broken reports
- Rebind reports
- Set a report theme
- Migration of report-level measures to the semantic model
- Tenant Settings tracking

Semantic Models

- Best Practice Analyzer
- Vertipaq Analyzer
- Semantic model edits (TOM)
- Metadata translations
- Semantic model refresh
- Visualize a refresh
- Semantic model backups
- Run DAX with impersonation
- Manage Query Scale Out

Direct Lake

- Migration to Direct Lake
- Check Direct Lake guardrails
- Warm the cache for Direct Lake
- Analyze Delta tables for Direct Lake
- Fallback to DirectQuery diagnostics
- Update connection of a Direct Lake semantic model

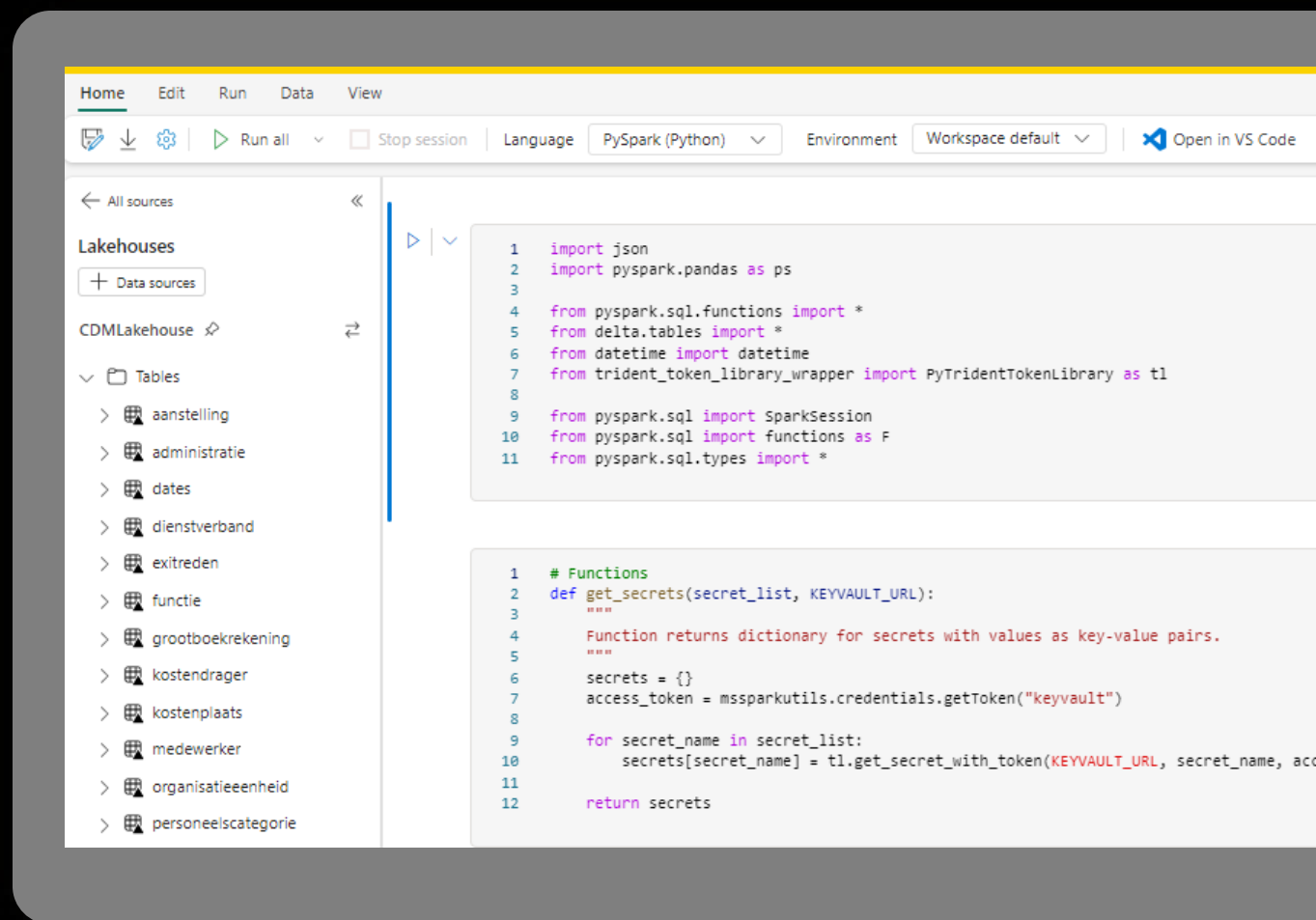
Capacities

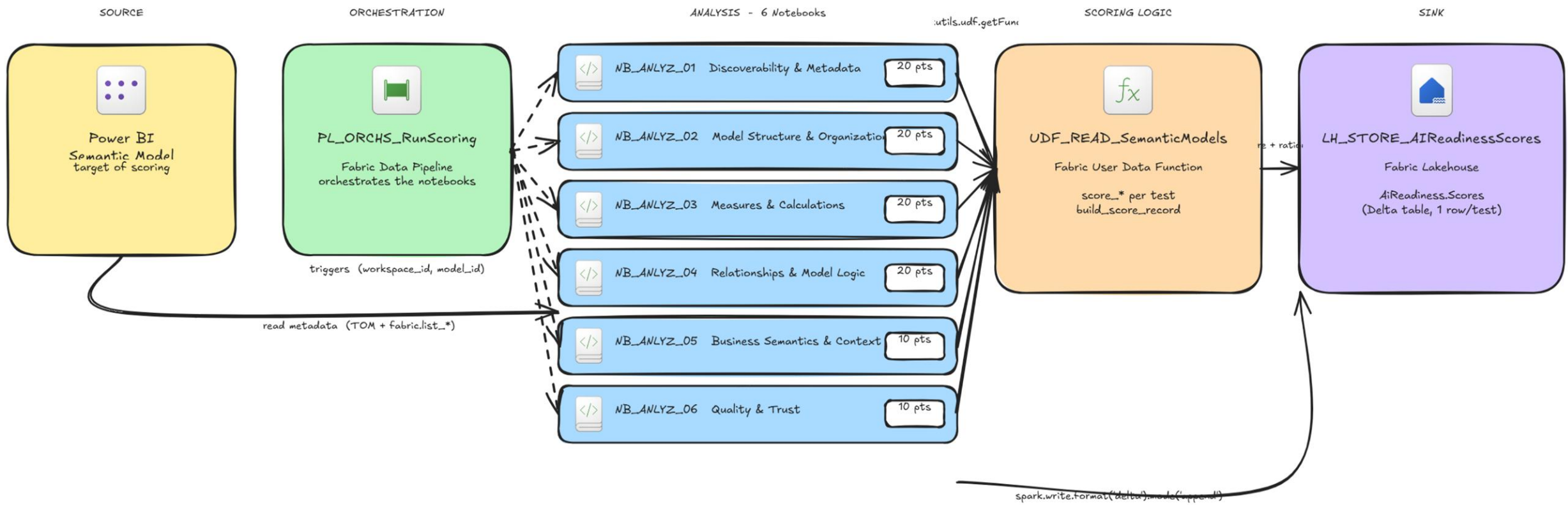
- Migration from P SKUs to F SKUs
- Migration from FT SKUs to F SKUs
- Capacity management

And more...

What are notebooks?

- Code first
- Web-based interface
- Cell based code blocks
- Runs on nodes (part of Fabric capacity)
- Often used languages are Python, Spark & Markdown
- Used by data engineers for data ingest, prep and transformations
- Used by data scientist for experiments and models





The architecture

Which components are part of this automated solution?





Monitoring and improving readiness

**Semantic models evolve over time,
and so should your AI readiness score**

Assessment integration

- Availability of Semantic Models (feature flag) for AI
- As part of regular governance / monitoring processes
- Automation of Copilot capacities

△ Approved for Copilot

Turn on this setting to approve your model for Copilot use. This improves visibility in search results and removes the warning in standalone Copilot in Power BI that your organization hasn't approved the data. [Learn more](#)

☒ Approved for Copilot

Apply

Discard

Solution open ends

- Scaling at enterprise level is not tested (yet)
 - Some technical validations should be done manually
(such as if descriptions make sense, currently it's only a check on existence of descriptions)
 - Capacity impact to run these assessments
 - Ever evolving AI landscape requires reevaluating the assessment over time
- ... you name it!

Recap & gotcha!

Semantic Models

- ... is the place where context is added to data!
- ... is where end-users will interact based out of Copilot and/or Data Agents

Reliable AI answers

- ... starts with adapting best practices in your semantic models
- ... can be measured programmatically
- ... all starts with the right mindset and governance!

**LET ME KNOW
WHAT YOU THINK!**

Measuring AI readiness

- ... is possible for sure on technical terms
- ... leaves room for quality checks (especially for context in AI notes & descriptions)
- ... should be regularly executed and reevaluated at scale!

Thanks for attending!



Marc Lelijveld

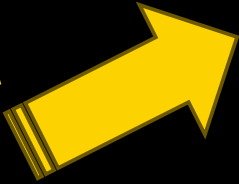
Architect & Technical Advocate
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 Data-Marc.com



**Session
Evals** 

These slides + the code

<https://github.com/marclelijveld/Slide-decks>

<https://github.com/marclelijveld/fabric-automation>